

Measurement (7 th std)

Choose the Appropriate Answer

1. The perimeter of a parallelogram whose adjacent sides are 6 cm and 5 cm is

(a) 12 cm (b) 10 cm (c) 24 cm (d) 22 cm

Ans: (d) 22 cm

$$\begin{aligned}\text{Perimeter of the parallelogram} &= 2 \times (b + h) \\ &= 2 \times (6 + 5) = 2 \times 11 = 22 \text{ cm}\end{aligned}$$

2. The area of a parallelogram whose base 10 m and height 7 m is

(a) 70 sq. m (b) 35 sq. m (c) 7 sq. m (d) 10 sq. m

Ans: (a) 70 sq.m

$$\begin{aligned}\text{Area of the parallelogram} &= \text{base} \times \text{height} \\ &= 10 \text{ m} \times 7 \text{ m} = 70 \text{ sq.m}\end{aligned}$$

3. The base of the parallelogram with area is 52 sq. cm and height 4 cm is

(a) 48 cm (b) 104 cm (c) 13 cm (d) 26 cm

Ans: (c) 13 cm

$$\text{Base} = \text{Area} / \text{Height} = 52 \text{ sq. cm} / 4 \text{ cm} = 13 \text{ cm}$$

4. What happens to the area of the parallelogram, if the base is increased 2 times and the height is halved?

(a) Decreases to half (b) Remains the same (c) Increases by two times (d) none

Ans: (b) Remains the same

5. In a parallelogram the base is three times its height. If the height is 8 cm then the area is

(a) 64 sq. cm (b) 192 sq. cm (c) 32 sq. cm (d) 72 sq. cm

Ans: (b) 192 sq. cm

6. The area of the rhombus with side 4 cm and height 3 cm is

(a) 7 sq. cm (b) 24 sq. cm (c) 12 sq. cm (d) 10 sq. cm

Ans: (c) 12 sq. cm

$$\text{Area} = \text{Base} \times \text{Height} = 4 \times 3 = 12 \text{ Sq. cm}$$

7. The area of the rhombus when both diagonals measuring 8 cm is

(a) 64 sq. cm (b) 32 sq. cm (c) 30 sq. cm (d) 16 sq. cm

Ans: (b) 32 sq. cm

$$\text{Area} = 1/2 (d_1 \times d_2) = 1/2 \times 8 \times 8 = 32 \text{ sq. cm}$$

8. The area of the rhombus is 128 sq. cm and the length of one diagonal is 32 cm. The length of the other diagonal is
(a) 12 cm (b) 8 cm (c) 4 cm (d) 20 cm

Ans: (b) 8 cm

$$\frac{1}{2} \times d_1 \times d_2 = 128 \Rightarrow d_2 = \frac{128 \times 2}{32} = 8 \text{ cm}$$

9. The height of the rhombus whose area 96 sq. m and side 24 m is
(a) 8 m (b) 10 m (c) 2 m (d) 4 m

Ans: (d) 4 m

10. The angle between the diagonals of a rhombus is
(a) 120° (b) 180° (c) 90° (d) 100°

Ans: (c) 90°

Angles of a rhombus bisect at right angles.

11. The area of the trapezium, if the parallel sides are measuring 8 cm and 10 cm and the height 5 cm is
(a) 45 sq. cm (b) 40 sq. cm (c) 18 sq. cm (d) 50 sq. cm

Ans: (a) 45 sq. cm

$$\frac{1}{2} \times h \times (a + b) = \frac{1}{2} \times 5 \times (10 + 8) = 45$$

12. In a trapezium if the sum of the parallel sides is 10 cm and the area is 140 sq. cm, then the height is
(a) 7 cm (b) 40 cm (c) 14 cm (d) 28 cm

Ans: (d) 28 cm

$$\text{Area} = \frac{1}{2} \times h \times (a + b) = 140 = \frac{1}{2} \times h \times 10 \Rightarrow h = 28$$

13. When the non-parallel sides of a trapezium are equal then it is known as
(a) a square (b) a rectangle (c) An isosceles trapezium (d) a Parallelogram

Ans: (c) an isosceles trapezium

14. The area of a rectangular sheet is 500 cm^2 . If the length of the sheet is 25 cm, what is its width?
(a) 20 cm (b) 17 cm (c) 30 cm (d) 25 cm

Ans: (a) 20 cm

$$\begin{aligned} \text{Area} &= \text{Length} \times \text{Width} \\ \text{Width} &= \frac{\text{Area}}{\text{Length}} = \frac{500 \text{ cm}^2}{25 \text{ cm}} = 20 \text{ cm} \end{aligned}$$

15. If the area of rectangle increases from 2 cm^2 to 4 cm^2 the perimeter will
(a) Increase (b) Decrease (c) Remains same (d) None of these

Ans: (a) Increase.

16. The area of a square whose perimeter is 4 m

- (a) 1 m^2 (b) 4 m^2 (c) 2 m^2 (d) 3 m^2

Ans: (a) 1 m^2

The perimeter of a square is given by the formula $P = 4s$

$$\text{Perimeter } 4 \text{ cm} \quad 4 = 4s \quad s = 4 / 4 = 1$$

$$\text{Area of the square} = a \times a = 1 \text{ sq. m}$$

17. The area of rectangle whose length is 15 cm & breadth is 6 m.

- (a) 9000 cm^2 (b) 90 cm^2 (c) 9 cm^2 (d) 900 cm^2

Ans: (a) 9000 cm^2

The area of a rectangle = length $\times$ breadth.

$$\text{Area} = 15 \text{ cm} \times 600 \text{ cm} = 9000 \text{ cm}^2.$$

18. The area of parallelogram is

- (a) Base + height (b) base $\times$ height (c) base $\times$ base (d) height $\times$ height

Ans: (b) base $\times$ height

19. If the area of the triangle is 36 cm^2 and the height is 3 cm, the base of the triangle will be

- (a) 12 cm (b) 39 cm (c) 108 cm (d) 24 cm

Ans: (d) 24 cm

$$\text{Area of a triangle} = \frac{1}{2} \times \text{base} \times \text{height}.$$

$$36 = \frac{1}{2} \times \text{base} \times 3.$$

$$\text{base} = \frac{2 \times 36}{3} = \frac{72}{3} = 24 \text{ cm}.$$

20. Which of the following has the formula : Base $\times$ Height

- (a) Area of parallelogram (b) Area of quadrilateral (c) Area of triangle
(d) Area of trapezium

Ans: (a) Area of parallelogram

21. The distance around a circular region is known as its

- (a) Area (b) Diameter of circle (c) Circumference (d) radius

Ans: (c) Circumference

22. The perimeter of square of side 2.5 m is

- (a) 10 m (b) 10 m^2 (c) 6.25 m^2 (d) 6.25 m

Ans: (b) 10 m^2

Perimeter of a square $P = 4 \times s$ ($s = 2.5$ m) $P = 4 \times 2.5$ m = 10 m.

23. The perimeter of rectangle of length 1.5 cm & breadth 2 cm is
 (a) 3.4 cm (b) 7 cm (c) 6 cm (d) 3.5 cm

Ans: (b) 7 cm

Perimeter of a rectangle is given by $P = 2(l + b)$, $P = 2(1.5 \text{ cm} + 2 \text{ cm})$. $P = 2 \times 3.5 \text{ cm} = 7 \text{ cm}$.

24. The area of parallelogram whose base 6 cm & altitude 7 cm is
 (a) 42cm^2 (b) 42 cm (c) 13 cm^2 (d) 13 cm

Ans: (a) 42cm^2

Area of the parallelogram = base $\times$ altitude. $\text{Area} = 6 \text{ cm} \times 7 \text{ cm} = 42 \text{ cm}^2$.

25. The height of parallelogram whose area is 35 cm^2 and altitude 7 cm
 (a) 5 cm (b) 5 cm^2 (c) 245 cm (d) 245 cm^2

Ans: (a) 5 cm

Area of a parallelogram = base $\times$ height. $35 \text{ cm}^2 = \text{base} \times 7 \text{ cm}$. $\text{base} = \frac{35 \text{ cm}^2}{7 \text{ cm}} = 5 \text{ cm}$.

26. Area of triangle whose base is 15 cm and corresponding altitude is 6 cm will be

- (a) 45 cm^2 (b) 90 cm^2 (c) 45 cm (d) 90 cm

Ans: (a) 45 cm^2

Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height}$. $\text{Area} = \frac{1}{2} \times 15 \text{ cm} \times 6 \text{ cm} = 45 \text{ cm}^2$.
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27. Find the area of a right triangle whose base is 3 cm, perpendicular is 2 cm and hypotenuse is 5 cm.

- (a) 3 cm^2 (b) 7.5 cm^2 (c) 5 cm^2 (d) 6 cm

Ans: (a) 3 cm^2

28. What will be the area of circular button of radius 7cm is

- (a) 154 cm^2 (b) 49 cm^2 (c) 154 cm (d) $3.14 \times 7 \text{ cm}^2$

Ans: (a) 154 cm^2

$$\text{Area of a circle} = \pi r^2$$

$$A = \frac{22}{7} \times (7 \text{ cm})^2 = 154 \text{ cm}^2$$

29. The perimeter of circle is its

- (a) Area (b) Circumference (c) Radius (d) Diameter

Ans: (b) Circumference

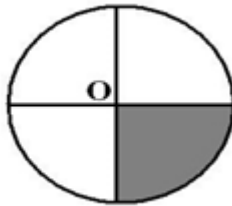
30. Diameter is _____.

- (a) Twice radius (b) half radius (c) equal to radius (d) one-third of radius

Ans: (a) Twice radius

31. If the area of circle is 44 cm^2 , the area of shaded portion will be

- (a) 11 cm^2 (b) 11 cm (c) 22 cm^2 (d) 22 cm^2



Ans: (a) 11 cm^2

32. If the radius of pipe is 1 cm, the circumference of pipe will be

- (a) 62.8 cm (b) 6.28 cm (c) 62.8 cm^2 (d) 6.28 cm

Ans: (b) 6.28 cm

$$\text{circumference of a circle is } C = 2\pi r. \quad (\pi = 3.14)$$

$$C = 2 \times \pi \times 1 \text{ cm.}$$

$$C \approx 2 \times 3.14159 \text{ cm} \approx 6.28 \text{ cm.}$$

33. The area of a circle is $49\pi \text{ cm}^2$. Its circumference is

- (a) $7\pi \text{ cm}$ (b) $14\pi \text{ cm}$ (c) $21\pi \text{ cm}$ (d) $28\pi \text{ cm}$

Ans: (b) $14\pi \text{ cm}$

$$\text{area of a circle} = \pi r^2 = 49\pi$$

$$r^2 = 49.$$

$$r = \sqrt{49} = 7 \text{ cm.}$$

$$\text{circumference of a circle is given by } C = 2\pi r = 2\pi(7) = 14\pi \text{ cm}$$

34. The circumference of two circles are in the ratio 2: 3. The ratio of their areas is

- (a) 2 : 3 (b) 4 : 9 (c) 9 : 4 (d) none of these

Ans: (b) 4: 9

35. The areas of the two circles are in the ratio 4: 9. The ratio of their circumference is

- (a) 2 : 3 (b) 4 : 9 (c) 9 : 4 (d) 4 : 9

Ans: (a) 2: 3

36. Find the circumference of a circle of diameter 21 cm.

- (a) 62 cm (b) 64 cm (c) 66 cm (d) 68 cm

Ans: (c) 66 cm

$$C = 22 \times \frac{21}{7} \text{ cm} = 22 \times 3 \text{ cm} = 66 \text{ cm.}$$

37. Find the area of a circle whose circumference is 52.8 cm.

- (a) 221.76 cm² (b) 220.76 cm² (c) 200.76 cm² (d) none of these.

Ans: (a) 221.76 cm²

circumference of a circle is $C = 2\pi r$

$$r = \frac{52.8 \times 7}{2 \times 22} = \frac{52.8 \times 7}{44} = \frac{369.6}{44} = 8.4 \text{ cm.}$$

$$\text{The area of the circle} = \pi r^2 = \frac{22}{7} \times (8.4)^2 = 221.76 \text{ cm}^2$$

38. The area of triangle is

- (a) Base x Height (b) $(1/2) \times (\text{Base} \times \text{Height})$ (c) $(1/2) \times (\text{Base} + \text{Height})$ (d) Base + Height

Ans: $(1/2) \times (\text{Base} \times \text{Height})$

39. The value of π (pi) is

- (a) Ratio of circumference to diameter (b) Diameter to circumference (c) 21 / 17 (d) 3.14

Ans: (d) 3.14

40. The circumference of the circle is

- (a) πr (b) πr^2 (c) $\pi \times 2r$ (d) $\pi + 2r$

Ans: (c) $\pi \times 2r$

41. The diameter of a circle is

- (a) r^2 (b) $2r$ (c) $2\pi r^3$ (d) πr^2

Ans: (b) $2r$

42. Which of the following is the example of the circle?

- (a) a chair (b) a bottle cap (c) a cup (d) a table

Ans: (b) a bottle cap

43. The area of a circle is

- (a) $2\pi r$ (b) $2\pi r^2$ (c) πr^2 (d) πd

Ans: (c) πr^2

44. The value of constant π is

- (a) 31.4 (b) $22/7$ (c) $7/22$ (d) 314

Ans: (b) $22/7$

45. Perimeter of a square

- (a) Side $\times$ Side (b) $3 \times$ Side (c) $4 \times$ Side (d) $2 \times$ Side

Ans: (c) $4 \times$ side

46. The perimeter of a rectangle whose length (l) and breadth (b) are given, is

- (a) $2(l + b)$ (b) $2l + b$ (c) $2l + 3b$ (d) None of these

Ans: (a) $2(l + b)$

47. Area of a square

- (a) side $\times$ side (b) $2 \times$ side (c) $3 \times$ side (d) $4 \times$ side

Ans: (a) side $\times$ side

48. Area of a rectangle of length l and breadth b is

- (a) $l \times b$ (b) $l + b$ (c) $2 \times (l + b)$ (d) $6 \times (l + b)$

Ans: (a) $l \times b$

49. Area of a triangle

- (a) base $\times$ height (b) $1/2 \times$ base $\times$ height (c) $1/3 \times$ base $\times$ height (d) $1/4 \times$ base $\times$ height

Ans: (b) $1/2 \times$ base $\times$ height

50. The area of a square plot is 1600 m^2 . The side of the plot is

- (a) 40 m (b) 80 m (c) 120 m (d) 160 m

Ans: (a) 40 m

$$\text{Side} = \sqrt{1600} = 40 \text{ m}$$

51. The area of a square is 121 m^2 . Its side is

- (a) 11 m (b) 21 m (c) 31 m (d) 41 m

Ans: (a) 11 m

$$\text{Side} = \sqrt{121} = 11 \text{ m}$$

52. The area of a rectangular field is 250 m^2 . If the breadth of the field is 10 m, find its length. (a) 25 m (b) 50 m (c) 100 m (d) 125 m

Ans: (a) 25 m

$$\text{Length} = 250/10 = 25 \text{ m}$$

53. The perimeter of a rectangle is 30 m. Its length is 10 m. Its breadth is

(a) 5 m (b) 10 m (c) 15 m (d) 3 m

Ans: (a) 5 m

perimeter of a rectangle $\Rightarrow 30 = 2 (\text{Length} + \text{Breadth})$

$$30 = 2 (10 + \text{Breadth}) \Rightarrow 30 / 2 = (10 + \text{breadth}) \Rightarrow \text{Breadth} = 5 \text{ m}$$

54. The perimeter of a square is 48 cm. Its area is

(a) 144 cm² (b) 12 cm² (c) 48 cm² (d) 100 cm²

Ans: (a) 144 cm²

$$\text{Side} = 48 / 4 = 12 \text{ cm} \Rightarrow \text{Area} = 12 \times 12 = 144 \text{ cm}^2$$

55. The area of a rectangular room is 150 m². If its breadth is 10 m, then find its length.

(a) 15 m (b) 25 m (c) 50 m (d) 55 m

Ans: (a) 15 m

$$\text{Length} = 150 / 10 = 15 \text{ m}$$

56. A rectangular wire of length 40 cm and breadth 20 cm is bent in the shape of a square. The side of the square is

(a) 10 cm (b) 20 cm (c) 30 cm (d) 40 cm

Ans: (c) 30 cm

$$\text{Side of square} = 2 (40 + 20)$$

$$4s = 120 \Rightarrow s = 120 / 4 = 30 \text{ cm}$$

57. The area of a parallelogram of base 5 cm and height 3.2 cm is

(a) 8 cm² (b) 12 cm² (c) 16 cm² (d) 20 cm²

Ans: (c) 16 cm²

$$\text{Area} = 5 \times 3.2 = 16 \text{ cm}^2$$

58. The area of the parallelogram ABCD in which AB = 6.2 cm and the perpendicular from C on AB is 5 cm is

(a) 30 cm² (b) 29 cm² (c) 28 cm² (d) 31 cm²

Ans: (d) 31 cm²

$$\text{Area} = 6.2 \times 5 = 31 \text{ cm}^2$$

59. One of the sides and the corresponding height of a parallelogram are 3 cm and 1 cm respectively. The area of the parallelogram is

(a) 1 cm² (b) 3 cm² (c) 6 cm² (d) 12 cm²

Ans: (b) 3 cm²

60. The area of a parallelogram is 20 cm² and the height is 2 cm. Find the corresponding base.

(a) 3 cm (b) 4 cm (c) 5 cm (d) 10 cm

Ans: (d) 10 cm

$$\text{Base} = 20 / 2 = 10 \text{ cm}$$

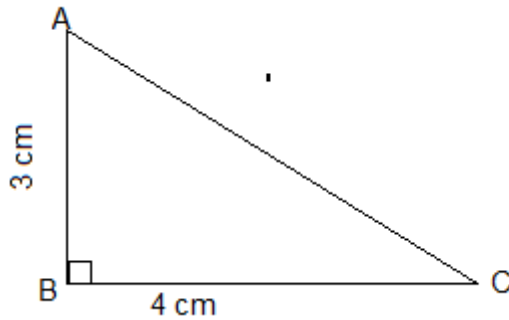
61. If the area of a parallelogram is 16 cm^2 and the base is 8 cm , find the height.

- (a) 1 cm (b) 2 cm (c) 3 cm (d) 4 cm

Ans: (b) 2 cm

$$\text{Height} = 16/8 = 2 \text{ cm}$$

62. Find the area of $\triangle ABC$



- (a) 3 cm^2
(b) 4 cm^2
(c) 6 cm^2
(d) 12 cm^2

Ans: (c) 6 cm^2

$$\begin{aligned} \text{Area of the triangle} &= (1/2) \times (\text{base} \times \text{height}) \\ &= (1/2) \times (3 \times 4) = (1/2) \times 12 = 6 \text{ cm}^2 \end{aligned}$$

63. The radius of a circle is 7 cm . Its circumference is

- (a) 22 cm (b) 44 cm (c) 11 cm (d) 66 cm

Ans: (b) 44 cm

$$\text{Circumference of the circle} = 2 \times \frac{22}{7} \times 7 = 44 \text{ cm}$$

64. The diameter of a circle is 28 cm . Find its circumference

- (a) 14 cm (b) 24 cm (c) 88 cm (d) 66 cm

Ans: (c) 88 cm

$$d = 28 \text{ cm} \Rightarrow r = 14 \text{ cm}$$

$$\text{Circumference of the circle} = 2 \times \frac{22}{7} \times 14 = 88 \text{ cm}$$

65. The radius of a circle is 7 cm . Find its area

- (a) 154 cm^2 (b) 77 cm^2 (c) 11 cm^2 (d) 22 cm^2

Ans: (a) 154 cm^2

$$\text{Area} = \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$$

66. When the circumference and area of a circle are numerically equal, what is the diameter numerically equal to?

- (a) Area (b) Circumference (c) 271 (d) 4

Ans: (d) 4

$$\text{Circumference of circle} = \text{Area of circle}$$

$$2\pi r = \pi r^2$$

$$d = \left(\frac{d}{2}\right)^2 \Rightarrow 4d = d^2 \Rightarrow d = 4$$

67. The area of the rhombus when both diagonals measuring 8 cm is
(a) 64 sq. cm (b) 32 sq. cm (c) 30 sq. cm (d) 16 sq. cm

Ans: **(b) 32 sq. cm**

$$\begin{aligned}\text{Area of the Rhombus} &= \frac{1}{2} \times d_1 \times d_2, \\ A &= \frac{1}{2} \times 8 \text{ cm} \times 8 \text{ cm} = \frac{1}{2} \times 64 \text{ cm}^2 = 32 \text{ cm}^2.\end{aligned}$$

68. In a parallelogram the base is three times its height. If the height is 8 cm then the area is

(a) 64 sq. cm (b) 192 sq. cm (c) 32 sq. cm (d) 72 sq. cm

Ans: **(b) 192 sq. cm**

$$\begin{aligned}b &= 3 \times h = 3 \times 8 \text{ cm} = 24 \text{ cm}. \\ \text{Area} &= b \times h = 24 \text{ cm} \times 8 \text{ cm} = 192 \text{ cm}^2\end{aligned}$$

69. The base of the parallelogram with area is 52 sq. cm and height 4 cm is

(a) 48 cm (b) 104 cm (c) 13 cm (d) 26 cm

Ans: **(c) 13 cm**

$$\begin{aligned}\text{Area of a parallelogram} &= \text{base} \times \text{height}. \\ 52 &= \text{base} \times 4. \\ \text{base} &= \frac{52}{4} = 13 \text{ cm}.\end{aligned}$$

70. The perimeter of a parallelogram whose adjacent sides are 6 cm and 5 cm is

(a) 12 cm (b) 10 cm (c) 24 cm (d) 22 cm

Ans: **(d) 22 cm**

$$\begin{aligned}\text{Perimeter of a parallelogram } P &= 2(a + b) \\ &= 2(6 \text{ cm} + 5 \text{ cm}) = 22 \text{ cm}\end{aligned}$$

71. When the non-parallel sides of a trapezium is referred as

(a) A square (b) Legs (c) An isosceles trapezium (d) A parallelogram

Ans: **(b) Legs**

72. In a trapezium if the sum of the parallel sides is 10 m and the area is 140 sq. m then the height is

(a) 7 cm (b) 40 cm (c) 14 cm (d) 28 cm

Ans: **(d) 28 m**

$$\begin{aligned}\text{Area of a trapezium} &= \frac{1}{2} \times (a + b) \times h, \\ 140 &= \frac{1}{2} \times 10 \times h, \\ h &= \frac{140}{5} \rightarrow 28 \text{ m}.\end{aligned}$$

73. The area of the rhombus with side 4 cm and height 3 cm is
 (a) 7 sq. cm (b) 24 sq. cm (c) 12 sq. cm (d) 10 sq. cm
Ans: (c) 12 sq. cm

$$\text{The area of a rhombus} = \text{base} \times \text{height}$$

$$\text{Area} = 4 \text{ cm} \times 3 \text{ cm} = 12 \text{ cm}^2$$

74. The height of the rhombus whose area 96 sq. m and side 24 m is
 (a) 8 m (b) 10 m (c) 2 m (d) 4 m
Ans: (d) 4 m

$$\text{Area of a rhombus} = b \times h,$$

$$96 = 24 \times h. \Rightarrow h = \frac{96}{24} = 4 \text{ m}$$

75. The area of parallelogram whose base 10 m and height 7 m is
 (a) 70 sq. m (b) 35 sq. m (c) 7 sq. m (d) 10 sq. m
Ans: (a) 70 Sq. m

$$\text{Area of a parallelogram} = \text{base} \times \text{height} = 10 \text{ m} \times 7 \text{ m} = 70 \text{ m}^2$$

76. The area of the rhombus is 128 sq. cm. and the length of one diagonal is 32 cm.
 The length of the other diagonal is
 (a) 12 cm (b) 8 cm (c) 4 cm (d) 20 cm
Ans: (b) 8 cm

$$\text{Area of a rhombus} = \frac{1}{2} d_1 d_2$$

$$128 = \frac{1}{2} \times 32 \times d_2.$$

$$d_2 = \frac{128}{16} = 8 \text{ cm}.$$

77. The area of the trapezium, if the parallel sides are measuring 8 cm and 10 cm
 and the height 5 cm is
 (a) 45 sq. cm (b) 40 sq. cm (c) 18 sq. cm (d) 50 sq. cm
Ans: (a) 45 sq. cm

$$\text{Area of a trapezium} = \frac{1}{2} \times (a + b) \times h$$

$$A = \frac{1}{2} \times 18 \text{ cm} \times 5 \text{ cm} = 45 \text{ cm}^2$$

78. What happens to the area of the parallelogram if the base is decreased 2 times
 and the height is halved?
 (a) Decreases to half (b) $(1/4)$ th of its original area (c) Increase by two
 times (d) None
Ans: $(1/4)$ th of its original area

79. What is the circumference of the circle of diameter 10 cm?
 (a) 35 cm (b) 30 cm (c) 31.4 cm (d) none of these
Ans: (c) 31.4 cm
80. The perimeter of the rectangular sheet is 100 cm. if the length is 35 cm, find its breadth?
 (a) 35 cm (b) $100 / 35$ cm (c) 15 cm (d) None of these
Ans: (c) 15 cm
81. If the radius of the circle is doubled, then the area is
 (a) Doubled (b) halved (c) 4 times (d) 3 times
Ans: (c) 4 times
82. The perimeter of the square and rectangle are same, then which figure has larger area
 (a) Square (b) Rectangle (c) Both (d) Can't say
Ans: (a) Square
83. The diagonal of a parallelogram divides it into two triangles with ----- area.
 (a) Equal (b) unequal (c) 3 times (d) 4 times
Ans: (a) Equal
84. Ram wants the four walls of this room painted green at of Rs. 5.60 per sq.m, if the dimensions of the rooms are 6m x 5m x 4m, then the cost of painting is
 (a) Rs. 493.00 (b) Rs. 492.85 (c) Rs. 492.80 (d) Rs. 495.00
Ans: (c) Rs. 492.80
85. If the radius of a bangle is 2 inches then find the diameter.
 (a) 3 inches (b) 4 inches (c) 5 inches (d) 6 inches
Ans: (b) 4 inches
86. The perimeter of the parallelogram is 48 cm, if one of the side measures 10 cm, the adjacent side would be,
 (a) 12 cm (b) 13 cm (c) 14 cm (d) 15 cm
Ans: (c) 14 cm
87. The three sides of the triangles are in the ratio 2 : 3 : 4 and the perimeter is 54 cm, then the sides (in cm) of triangles are
 (a) 12, 18, 24 (b) 13, 15, 26 (c) 10, 20, 24 (d) None
Ans: (a) 12, 18, 24
88. If the radius of the circle is $\pi / 8$ cm, find its circumference
 (a) $\pi / 4$ (b) $2 \pi / 16$ (c) $\pi^2 / 4$ (d) $2 \pi^2$
Ans: (c) $\pi^2 / 4$
89. Area of the trapezium is
 (a) $l \times b$ (b) $2 h (l + b)$ (c) $(1 / 2) \times (b \times h)$ (d) $2 \times d$ (sum of parallel angles)
Ans: (b) $2 h (l + b)$
90. Area between two concentric circles are
 (a) $(22 / 7) r$ (b) πr^2 (c) $A = \pi (R + r) (R - r)$ (d) $\pi r^2 h$
Ans: (c) $A = \pi (R + r) (R - r)$

91. In the formula $C = 2\pi r$, 'r' refers to
 (a) Circumference (b) Area (c) Rotation (d) Radius
Ans: (d) Radius

92. The distance around a closed figure is called its:
 (a) Area (b) Volume (c) Perimeter (d) Circumference
Ans: (c) Perimeter

93. If the circumference of the circle is 82π , the value of r is
 (a) 41 cm (b) 82 cm (c) 21 cm (d) 20 cm
Ans: (a) 41 cm

94. The ratio of the area of a circle to the area of its semi-circle is
 (a) 3:4 (b) 2: 1 (c) 1: 3 (d) 2:3
Ans: (b) 2: 1

$$\text{The ratio of the area of a circle to the area of its semi-circle} = \frac{A_{\text{circle}}}{A_{\text{semi-circle}}} = \frac{\pi r^2}{\frac{1}{2} \pi r^2} = 2 : 1$$

95. Area of the circular path is
 (a) $\pi (R^2 - r^2)$ sq. units (b) πr^2 sq. units (c) $2\pi r^2$ sq. units (d) $\pi r^2 + 2r$
Ans: (a) $\pi (R^2 - r^2)$ sq. units

96. Circumference of a circle is always
 (a) More than three times of its diameter (b) Three times of its diameter
 (c) Less than three times of its diameter (d) Three times of its radius
Ans: (a) More than three times of its diameter

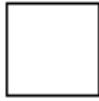
$$\begin{aligned} \text{Circumference of a circle} &= 2\pi r \quad (\text{Value of } \pi = 3.14) \\ &= 2 \times 3.14 \times r \Rightarrow (d = 2r) \\ \text{Circumference of a circle} &= 3.14 d \\ \text{Therefore, the circumference is always more than three} \\ &\text{times of its diameter.} \end{aligned}$$

97. In reference to a circle the value of π is equal to
 (a) Area / Circumference (b) Area / Diameter (c) Circumference / Diameter
 (d) Circumference / Radius
Ans: (c) Circumference / Diameter

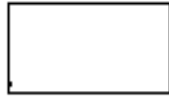
$$\begin{aligned} \text{Circumference of a circle} &= 2\pi r \Rightarrow d = 2r \\ \text{Circumference} &= \pi \times \text{Diameter} \\ \pi &= \text{Circumference/Diameter} \end{aligned}$$

98. The unit of area is always:
 (a) Linear (b) Square units (c) Cubic units (d) Ratio
Ans: (b) Square units

99. Which of the following figures has a circular shape?



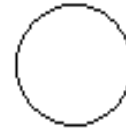
(a) Square



(b) Rectangle



(c) Triangle



(d) Circle

Ans: (d) Circle

100. Area of the square is 100 cm^2 and the area of the circle is 38.5 cm^2 . Find the area of the shaded path?



10 cm

- (a) 61.5 cm^2 (b) 73.5 cm^2 (c) 59.5 cm^2 (d) 64.5 cm^2

Ans: (a) 61.5 cm^2

Area of the square = $a \times a = 100 \text{ cm}^2$ Area of the circle = $\pi \times 3.5 \times 3.5 = 38.5 \text{ cm}^2$ Area of the shaded pathway = $100 - 38.5 = 61.5 \text{ cm}^2$
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